



Optimizing Pressure Swing Distillation for Di-n-Propyl Ether and n-Propyl Alcohol Separation Using Aspen HYSYS and Machine Learning Algorithms

Milad Karsaz¹ · Borhan Pourtalebi¹ · S. Majid Abdoli^{1,2} · Amir Raoof²

Received: 17 June 2025 / Revised: 9 November 2025 / Accepted: 4 January 2026 / Published online: 2 February 2026
© The Author(s) 2026

Abstract

This paper presents the design and simulation of a pressure-swing distillation (PSD) process for separating and purifying di-n-propyl ether (DnPE) from n-propyl alcohol (nPA) using Aspen HYSYS software. The minimum-boiling-point azeotrope formed at atmospheric pressure makes conventional separation methods ineffective. Three critical parameters, feed stage, feed temperature, and reflux ratio, are systematically optimized to minimize energy consumption. The optimized process achieves product purities of 99.5% DnPE and 98.7% nPA while reducing energy consumption by 15% compared to conventional distillation. Additionally, an XGBoost regression model is developed to predict reboiler heat duty with 95% accuracy, further enhancing process efficiency. Particle swarm optimization is employed to identify optimal operating conditions based on the machine learning predictions. This integrated computational approach demonstrates significant improvements in separation efficiency, highlighting the industrial potential of the optimized PSD process for azeotropic mixtures.

Keywords Azeotropic mixture · Pressure-swing distillation · Aspen HYSYS · Machine learning · Process optimization

Introduction

Di-n-propyl ether (DnPE) and n-propyl alcohol (nPA) are versatile organic compounds with extensive applications in chemical industries [1]. DnPE can be synthesized through the dehydration of nPA in the presence of sulfuric acid [2]. However, these compounds form a minimum boiling point azeotrope at atmospheric pressure, making conventional purification processes inefficient [3, 4]. Among various separation techniques, extractive distillation (ED) [5, 6] and pressure-swing distillation (PSD) have emerged as the most effective methods for azeotrope separation [7, 8].

Previous research has established the economic viability of PSD for DnPE/nPA separation. Lladosa et al. [9] demonstrated through Aspen HYSYS simulations that PSD offers greater cost-effectiveness compared to ED for this binary mixture. Similar advantages have been reported for other azeotropic systems. Zhang et al. [10] found that partially heat-integrated triple-column PSD processes provided cost-effective separation of acetonitrile, methanol, and benzene mixtures. For ethylene glycol and 1,2-butanediol mixtures, Mao et al. [11] reported that modified PSD processes with heat integration substantially reduced both CO₂ emissions and total annual costs. Additionally, Cui et al. [12] determined that PSD configurations were economically and energetically superior to heterogeneous azeotropic distillation for recovering benzene and isopropanol from wastewater.

Computer simulation represents a powerful approach for process design and optimization, particularly valuable given the time and cost constraints of experimental studies [13]. This research utilizes Aspen HYSYS V.11 to model continuous PSD for DnPE/nPA azeotrope separation, examining how column configuration and operational variables

✉ S. Majid Abdoli
s.abdoli1@uu.nl

¹ Chemical Engineering Faculty, Sahand University of Technology, Sahand New Town, P.O. Box 51335-1996, Tabriz, Iran

² Department of Earth Sciences, Utrecht University, Utrecht, The Netherlands

affect separation efficiency. The most economical design was selected for industrial implementation.

Machine learning has revolutionized the development of regression models as a significant subdomain of artificial intelligence [14, 15]. Its application to chemical unit operations design has shown tremendous promise, particularly in separation processes [16, 17]. While machine learning models require substantial data, they significantly reduce experimental workloads once developed. Furthermore, they enable rapid predictions and provide valuable insights into parameter contributions [18]. This study specifically focuses on predicting reboiler heat duty with emphasis on model speed and accuracy.

Recent research has successfully integrated machine learning with Aspen HYSYS to enhance process optimization [19]. Osman et al. [20] demonstrated how artificial neural networks improve product quality and reduce operational constraints in distillation columns. Qadeer et al. [21] developed a highly accurate random forest algorithm for determining relative humidity in energy systems. Additionally, Zhu et al. [22] introduced an innovative machine learning-based adaptive control method using dual RBFNNs, achieving superior performance in distillation column simulations. These studies collectively highlight the growing importance of combining machine learning with simulation tools in process engineering applications.

This research addresses the separation challenges of DnPE/nPA azeotropic mixtures through a two-phase approach. First, we train an XGBoost regression model using heat duty data collected across various operating conditions to evaluate its predictive capabilities. Second, we apply Particle swarm optimization to identify optimal operating conditions based on the XGBoost model. This integrated approach allows us to determine the relative importance of operating conditions and ultimately enhance separation efficiency.

Methodology

Simulation Procedure

The pressure-swing distillation process was simulated using the UNIQUAC (UNIversal QUAsiChemical) thermodynamic model, which has been validated for this system by Lladosa et al. [9, 23, 24]. The feed consisted of an equimolar binary mixture of DnPE and nPA with a mass flow rate of 1500 kg/h (approximately 18.49 kmol/h) at an initial temperature of 30 °C [9]. This thermodynamic model was selected for its proven accuracy in representing the vapor–liquid equilibrium behavior of the DnPE/nPA system under various operating conditions.

Machine Learning Data Preparation

For machine learning analysis, the preliminary dataset was precisely prepared from the data on pressure swing distillation originally published by Lladosa et al. [4]. This initial dataset underwent a crucial optimization step using Aspen HYSYS to refine the process parameters and outcomes. The distillation process performance depends on numerous parameters, including feed stage number, total number of stages, feed temperature, pressure, and reflux ratio. For this investigation, four key adjustable parameters were selected:

1. **Feed stage number:** varied across ten possible positions ranging from stage 2 to 11;
2. **Number of stages:** seven configuration options ranging from 10 to 16 total stages;
3. **Feed temperature:** varied within the range from 20 °C to 120 °C.
4. **Reflux ratio:** adjusted between 0.7 and 2.5.

These parameters were systematically varied to generate comprehensive datasets that captured the relationship between operating conditions and separation performance. The selection of these parameter ranges was based on preliminary simulations and practical operational constraints.

Data Standardization and Feature Construction

The machine learning analysis utilized data from 105 observations for the low-pressure (LP) column and 110 observations for the high-pressure (HP) column, each recording reboiler heat duty under different operating conditions. This dataset was divided into training (80%) and testing (20%) sets to ensure robust model development and evaluation.

Prior to model training, an intensive preprocessing step was implemented to enhance feature interpretability. Since the features (feed stage, number of stages, temperature, and reflux ratio) exhibited significantly different scales, standardization was essential to prevent high-magnitude features from dominating the model. Data standardization was performed according to Eq. (1) [11, 25]:

$$X = \frac{x - \bar{x}}{\sigma} \quad (1)$$

Here, x was the values of each feature, $\bar{x}$ was the mean value of x , and σ denoted the standard deviation of the feature. The data were standardized to have mean x and standard deviation σ for each feature.

To enhance model generalizability and prevent overfitting, cross-validation techniques were employed during the training process [25]. This approach allowed for more

robust performance assessment by evaluating the model on multiple subsets of the training data, thereby reducing the risk of developing a model that performs well only on specific data patterns.

XGBoost Algorithms for Model Training

XGBoost is an advanced gradient-boosting algorithm widely applied to regression, classification, and ranking problems [26]. Its regression capabilities make it particularly valuable for determining the relative significance of operating conditions in process optimization. The XGBoost algorithm from Scikit-Learn includes several critical hyperparameters: the number of trees, maximum tree depth, learning rate, and subsample ratio. Before implementing the model, these hyperparameters were meticulously tuned to optimize performance [27].

XGBoost employs the gradient boosting framework to build ensemble models through sequential decision tree construction. Unlike bagging methods that build trees independently, gradient boosting creates trees iteratively, with each new tree correcting errors made by previous trees. The final predictive model combines these individual trees to form a powerful ensemble. For regression applications, the predicted value for each data point can be expressed by Eq. (2) as a weighted sum of predictions from all constituent trees [18]:

$$\hat{y}(x_i) = \sum_{n=1}^n \omega_n h_n(x_i) \quad (2)$$

Here, $\hat{y}_{(x_i)}$ is the predicted value for the i^{th} data point; n is the number of trees; ω_n refers to the weight of the n^{th} tree; $h_n(x_i)$ refers to the n^{th} tree prediction for the i^{th} data point. Noting this, particularly in XGBoost, the prediction for every tree is dependent upon the sum of the weights for leaves of the path traversed by the data point.

PSO Optimization Method

Particle swarm optimization (PSO) is a powerful global stochastic optimization technique inspired by the collective behavior observed in bird flocking and fish schooling [28]. In PSO, each particle represents a candidate solution that explores the search space to find an optimal solution. Particles move according to their own experience and that of their neighbors, with movement guided by the best solutions discovered so far. Throughout the iterative process, particle velocities and positions are continuously adjusted to facilitate convergence toward global optimization.

PSO effectively balances exploration of new regions and exploitation of promising areas in the solution space. Its simple design and implementation make it particularly suitable for complex optimization problems, as it can avoid becoming trapped in local minima [29]. These characteristics make PSO an ideal choice for optimizing the XGBoost model parameters and operating conditions in the pressure-swing distillation process.

Pressure Swing Distillation

Pressure swing distillation (PSD) represents the most effective method for separating pressure-sensitive azeotropes [30, 31]. Unlike extractive distillation (ED) [32], PSD does not require additional components to achieve high-purity product separation [33, 34]. The separation is accomplished by manipulating the pressure across two or more distillation columns [35].

Figure 1 illustrates the vapor–liquid equilibrium (Y–X) diagram for the DnPE/nPA system at three different pressures. The diagram reveals that changing the composition of DnPE from 78 to 65 mol% under varying pressure conditions creates the thermodynamic possibility for effective separation using the PSD process. This pressure sensitivity of the azeotropic composition provides the fundamental basis for implementing pressure swing distillation as a viable separation technique for this system.

The operating pressures for the two distillation columns are carefully selected based on their effects on azeotropic composition and reboiler temperatures. A significant

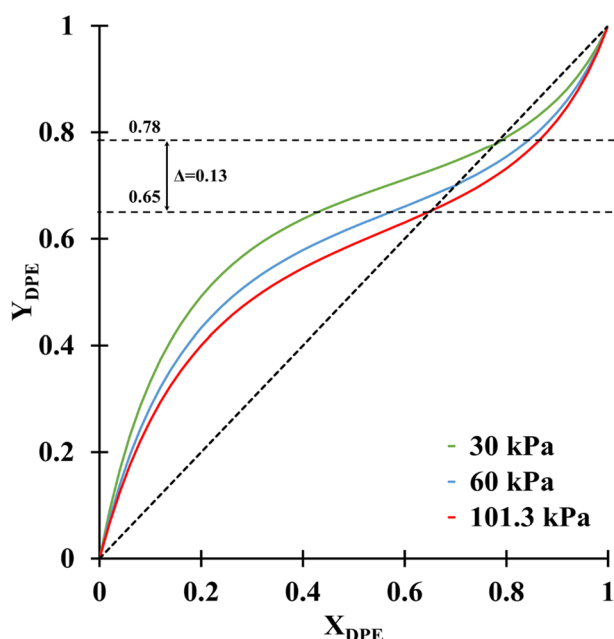


Fig. 1 Pseudo-binary Y–X diagram for the DPE/PA system at different pressures

Fig. 2 The pressure swing distillation process flow diagram for separation of DPE and PA

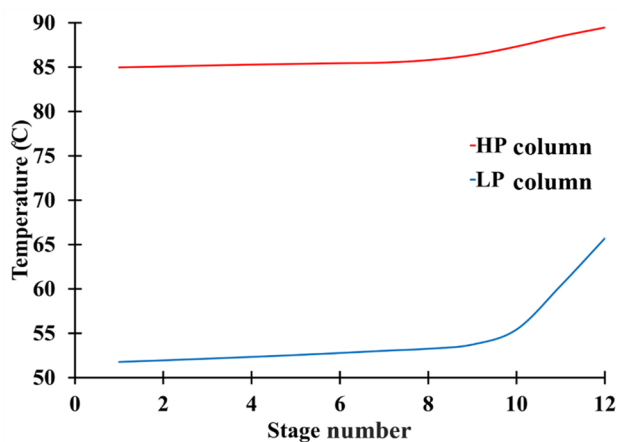
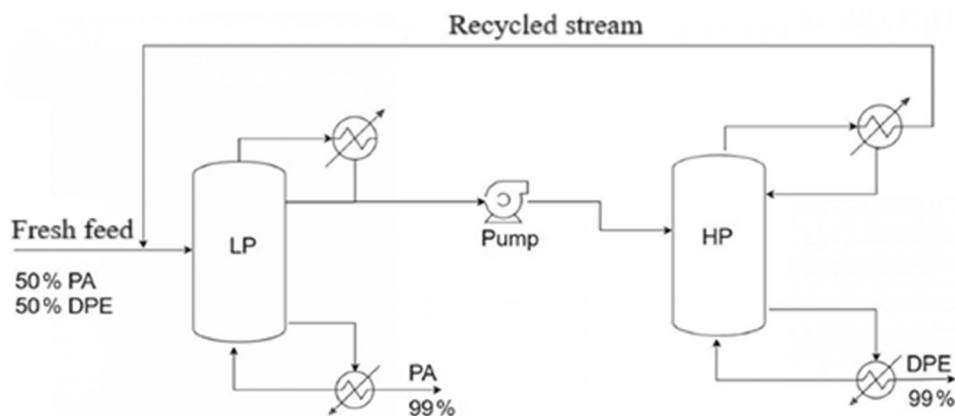


Fig. 3 Temperature profiles of the LP and HP columns

pressure difference between columns creates a larger shift in azeotropic composition, which reduces recycle flow rates and consequently lowers energy consumption. For the low-pressure (LP) column, the pressure must be high enough to maintain adequate reflux temperature. Meanwhile, the high-pressure (HP) column operating pressure should create at least a 5% shift in azeotropic composition over a moderate pressure range [36]. Based on these considerations,

pressures of 30 kPa and 101.325 kPa were selected for the LP and HP columns, respectively.

Figure 2 presents the process flow diagram for the implemented PSD system. An equimolar mixture of DnPE and nPA enters the LP column at the eighth stage. High-purity products are obtained as bottom streams from both columns—99 mol% nPA from the LP column and 99 mol% DnPE from the HP column. The LP column overhead stream is directed to the HP column at the fifth stage, while the HP column overhead product completes the cycle by returning to the LP column.

The optimal number of stages for both columns was determined using Aspen HYSYS short-cut distillation column analysis tool. Figures 3 and 4 illustrate the temperature and composition profiles throughout both columns, providing insights into separation performance and efficiency.

Temperature control tray selection follows the slope criterion method [37, 38], which identifies trays exhibiting significant temperature gradients from one tray to the next. This approach ensures effective temperature control and process stability. As shown in Fig. 3, tray 9 was selected for temperature control in the LP column, while tray 10 serves this function in the HP column. The temperature at these strategic locations is regulated by adjusting the reboiler heat

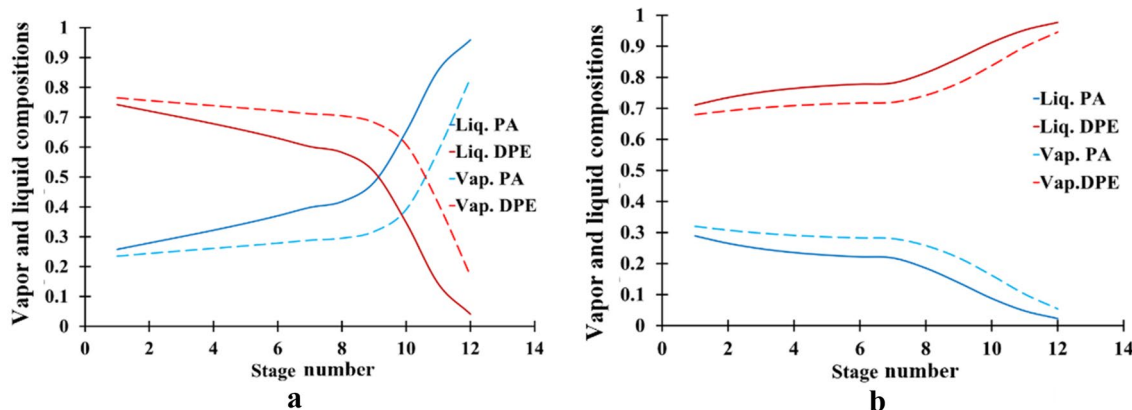


Fig. 4 Vapor and liquid compositions in the a LP and b HP

duties of their respective columns, allowing for responsive process control and optimization.

The temperature control trays were identified using the slope criterion method, which selects the tray exhibiting the largest temperature change between two adjacent trays. This criterion ensures that the chosen tray is located within the most thermodynamically sensitive region of the column, where small composition variations lead to measurable temperature responses.

As shown in Fig. 3, the LP column exhibits its maximum temperature gradient between trays 8 and 9 ($\Delta T \approx 2.6^\circ\text{C}$), while the HP column shows a similar gradient between trays 9 and 10 ($\Delta T \approx 2.5^\circ\text{C}$). These positions correspond to the intermediate region separating the rectifying and stripping sections, where mass and heat transfer rates are highest. Therefore, trays 9 (LP) and 10 (HP) were selected as optimal control points for maintaining process stability and energy efficiency.

Reproducibility and Numerical Error Evaluation

Although the present study is simulation-based, reproducibility of results was systematically verified to ensure numerical stability. Each Aspen HYSYS simulation was executed in triplicate under identical boundary and initialization conditions. The convergence criteria for material and energy balances were set to 1×10^{-6} , ensuring consistent steady-state solutions. The variation in reboiler heat duty and product purity among repeated runs was below 1%, confirming excellent numerical reproducibility.

For the machine learning stage, random seed control and five-fold cross-validation were employed to assess the reproducibility and generalization of the XGBoost regression model. The small standard deviation (<0.005) in R^2 and MSE across folds further verified the statistical reliability of the predictive framework. These systematic checks guarantee that all reported findings can be replicated following the described procedures.

Results and Discussion

A comprehensive sensitivity analysis was conducted to determine optimal PSD process conditions using Aspen HYSYS V.11 software. The initial configuration established by the short-cut design tool indicated that both the low-pressure (LP) and high-pressure (HP) columns should have 12 theoretical trays, with feed entry at tray 7 for the LP column and tray 6 for the HP column. Throughout the optimization process, several key parameters were systematically investigated to minimize reboiler heat duty (RHD): number of stages, feed stage location, feed temperature, and

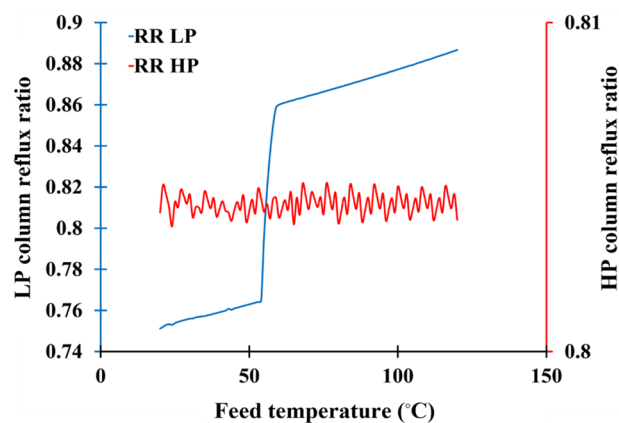


Fig. 5 Variation of LP and HP columns reflux ratio as a function of feed temperature

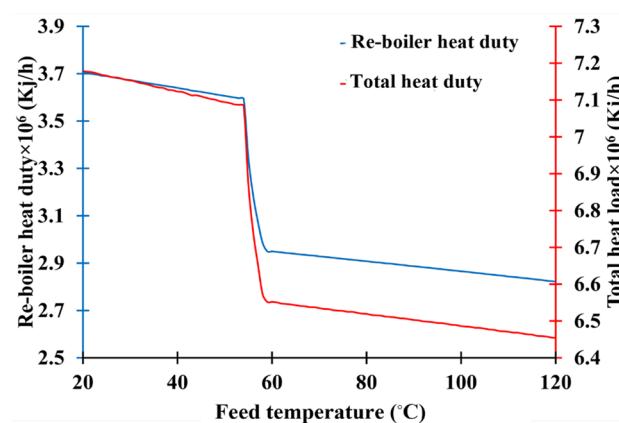


Fig. 6 Variation of total heat load and reboiler heat duty as a function of feed temperature

reflux ratio. Product purity was maintained at 99 mol% for all parametric variations to ensure fair comparison of energy requirements.

The reboiler duty decreases steadily with additional stages until approximately 12 stages, after which the energy savings become marginal. Based on this diminishing return, 12 theoretical stages were selected as the optimal configuration for both columns, balancing separation efficiency with equipment costs.

Feed temperature effects on system performance were investigated by varying this parameter from 20°C to 120°C while monitoring column reflux ratios, overall heat load, and reboiler duty. Figures 5 and 6 reveal approximately linear relationships within the temperature ranges of 20 – 54°C and 59 – 120°C , with a notable discontinuity between 54°C and 59°C . This inflection point corresponds to a phase change in the feed stream, causing the overall heat load and reboiler duty to decrease as temperature increases beyond this point, while simultaneously increasing the reflux ratio. Interestingly, the HP column reflux ratio showed minimal

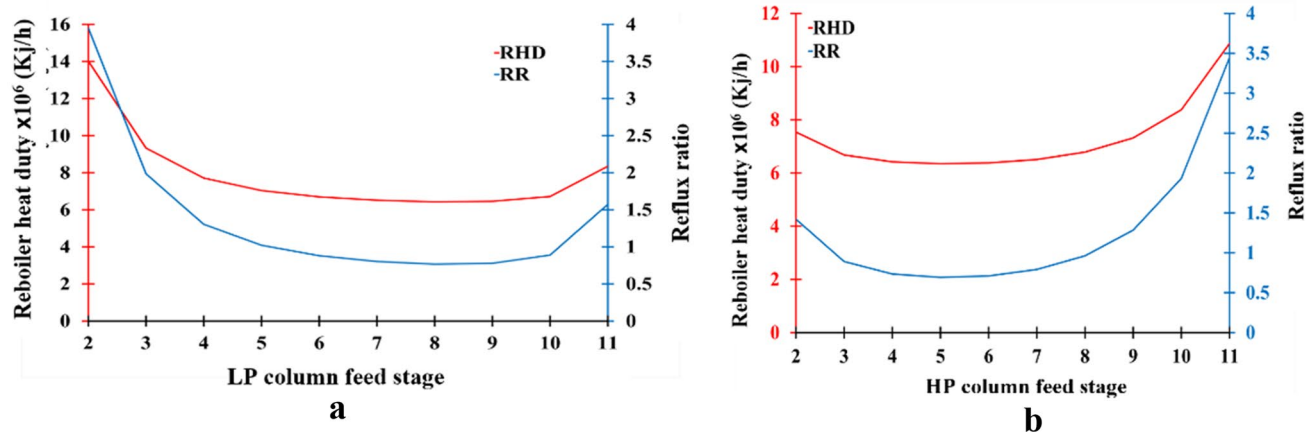


Fig. 7 Variation of total heat duty and reflux ratio as a function of feed stage, (a) LP Column, and (b) HP Column

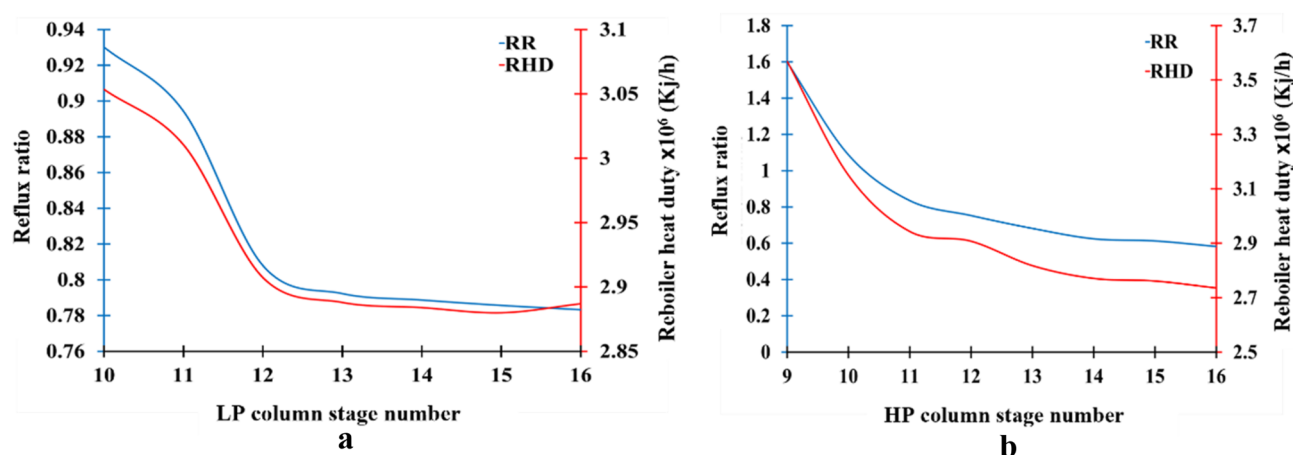


Fig. 8 Variation of reflux ratio and total reboiler heat duty as a function of stage number for the LPC (a) and HPC (b)

variation with feed temperature, exhibiting only small-amplitude fluctuations that can be considered negligible. This stability is attributed to the relatively consistent state of the HP column feed stream across the temperature range studied.

As illustrated in Fig. 5, the reflux ratio of the HP column exhibits only minor variations with increasing feed temperature. This behavior arises from the reduced temperature sensitivity of vapor–liquid equilibrium at elevated pressures. At 101.325 kPa, the DnPE/nPA system experiences lower relative volatility and a flatter equilibrium curve compared to the LP column operating at 30 kPa. Consequently, moderate temperature changes in the feed stream cause negligible alterations in vapor composition and condensation behavior, resulting in nearly constant reflux ratio values.

This observation indicates that the HP column operates within a thermodynamically stable region where the liquid–vapor phase balance is less affected by feed thermal perturbations. Such stability is advantageous for maintaining

consistent separation efficiency under minor process disturbances.

Figure 7 demonstrates how the total reboiler heat duty and reflux ratio vary with feed stage location for both columns. The data reveals a characteristic U-shaped curve for reboiler duty as the feed stage increases from 2 to 11. The minimum energy consumption occurs at stages 8 and 5 for the LP and HP columns, respectively, identifying these as the optimal feed entry positions. This optimization of feed location significantly impacts overall process efficiency by minimizing energy requirements while maintaining separation performance.

The relationship between the number of theoretical stages, reflux ratio, and reboiler heat duty is illustrated in Fig. 8. As the number of trays in each column increased from 10 to 16, both the required reflux ratio and reboiler heat duty decreased notably. This inverse relationship occurs because fewer stages necessitate higher reflux ratios to achieve the same separation, resulting in increased energy consumption and larger column diameters.

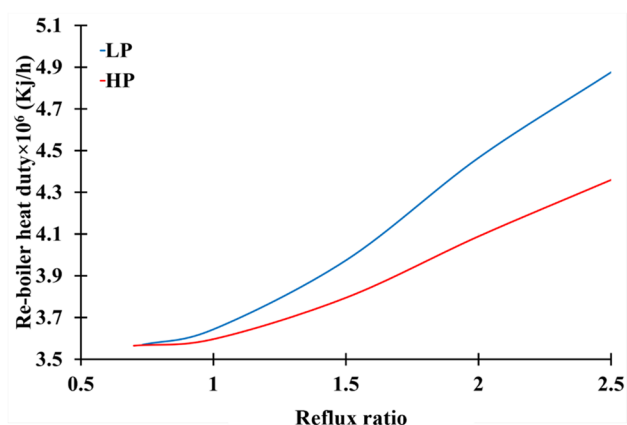


Fig. 9 Variation of the reboiler heat of LP and HP columns duty as a function reflux

Reflux ratio optimization is crucial for process profitability. To investigate this relationship, the reflux ratios for both LP and HP columns were systematically varied from 0.7 to 2.5, with the resulting effects on reboiler heat duty

illustrated in Fig. 9. The analysis revealed a strong correlation between reflux ratio and energy consumption, with reboiler heat duty increasing proportionally with higher reflux ratios. Minimum energy consumption was achieved at reflux ratios of 0.73 and 0.7 for the LP and HP columns, respectively. These optimal values represent the balance point where sufficient separation is maintained while minimizing energy requirements.

The XGBoost regression model performance was optimized through careful tuning of four critical hyperparameters: number of trees, maximum tree depth, learning rate, and subsampling rate. Model adequacy was rigorously assessed using coefficient of determination (R^2) and mean squared error (MSE) metrics [39]. Figures 10 and 11 present the hyperparameter tuning results for the LP and HP columns, respectively.

The effect of the number of estimators ($n_estimators$) on model performance is shown in Figs. 10a and 11a. Model accuracy improved with increasing $n_estimators$ until reaching a plateau beyond which additional trees provided

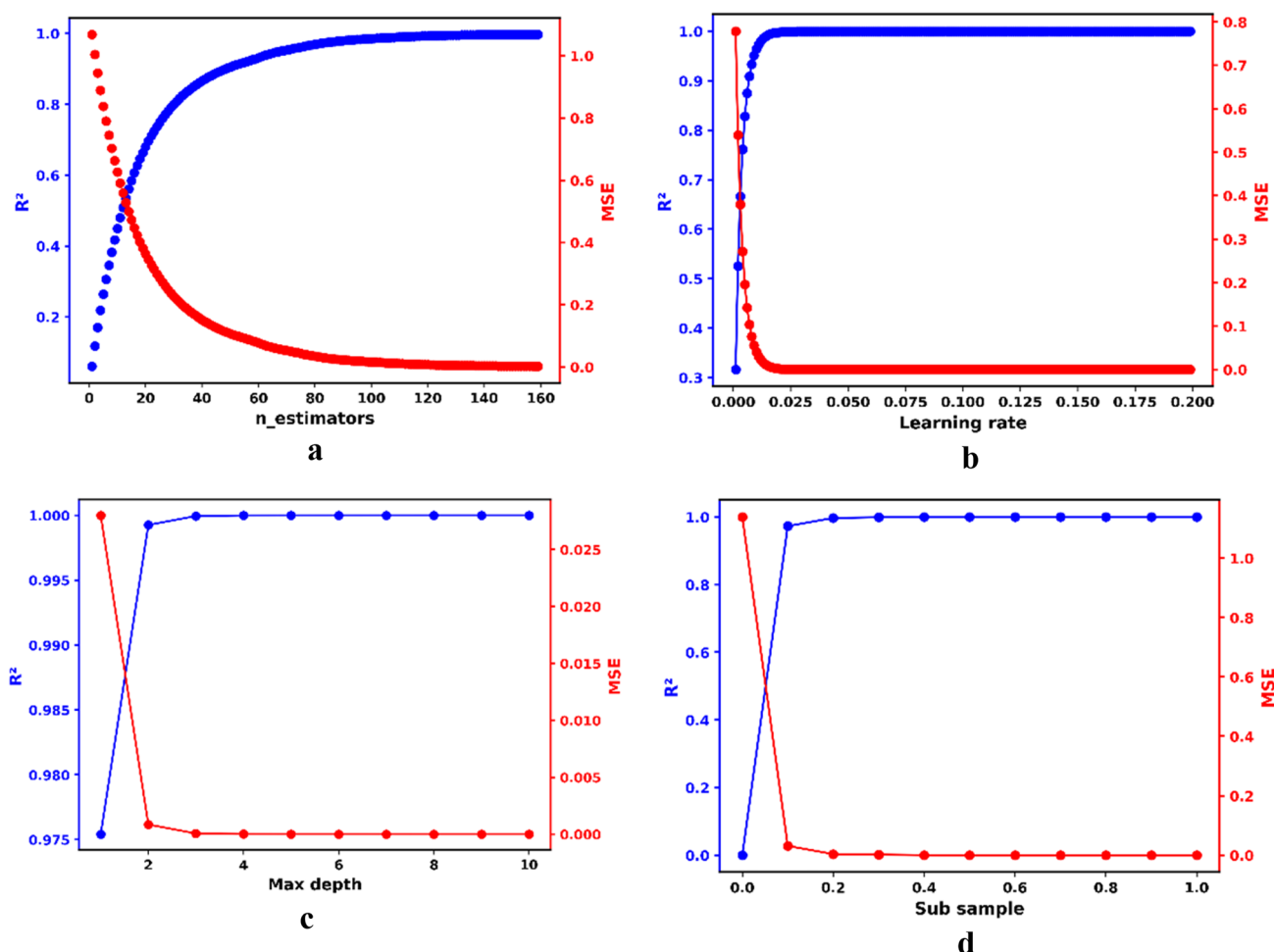


Fig. 10 The impact of the (a) number of trees, (b) learning rate, (c) maximum tree depth, and (d) subsample on the coefficient of determination (R^2) and mean squared error (MSE) of an XGBoost regression model for LP

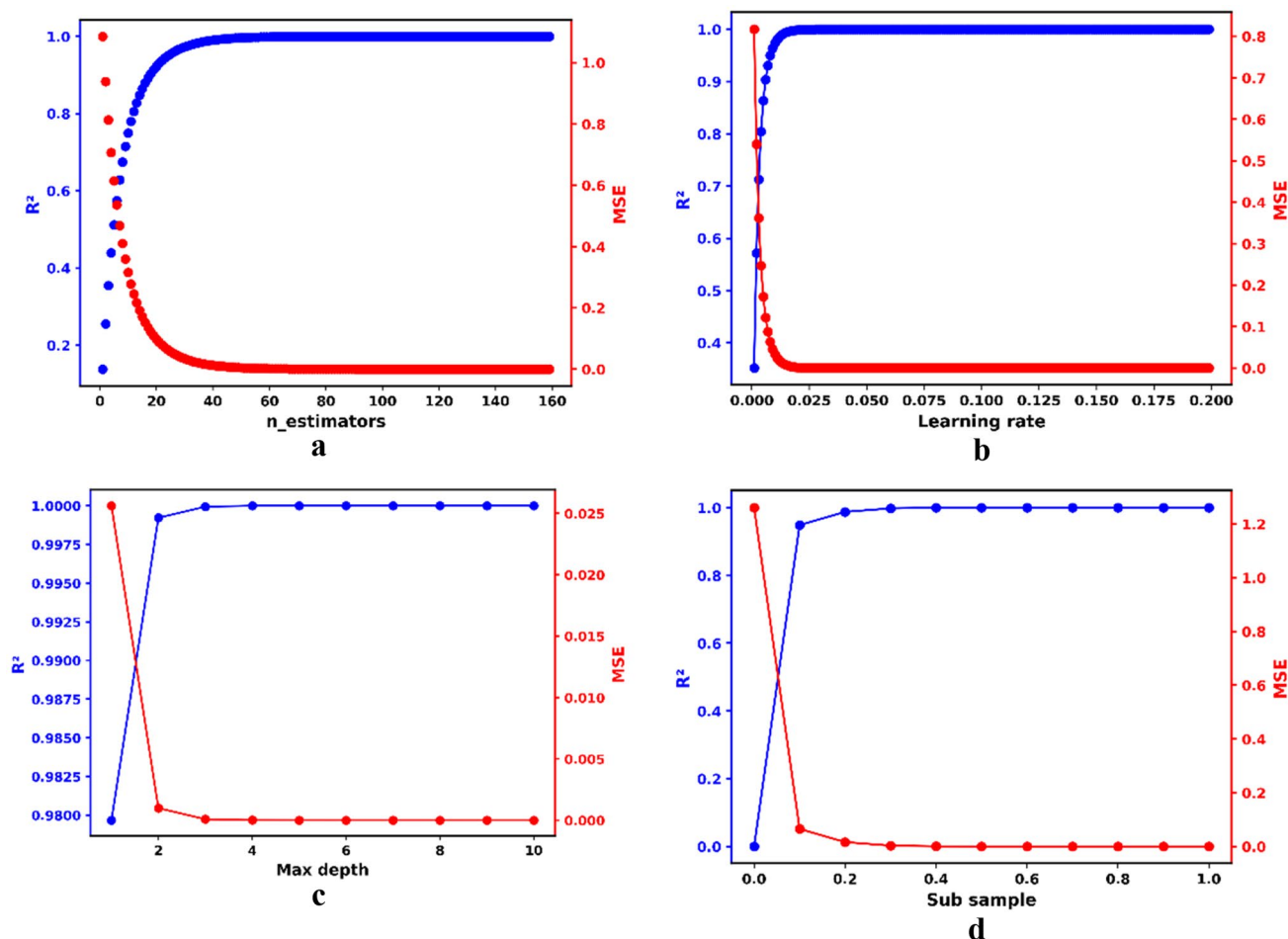


Fig. 11 The impact of the (a) number of trees, (b) learning rate, (c) maximum tree depth, and (d) subsample on the coefficient of determination (R^2) and mean squared error (MSE) of an XGBoost regression model for HP

diminishing returns while increasing computational cost. The optimal values were determined to be 100 trees for the LP column model and 60 trees for the HP column model, yielding impressive performance metrics: an MSE of 0.00157 and R^2 of 0.9776 for the LP model, and an MSE of 0.00101 and R^2 of 0.9842 for the HP model. With the number of trees fixed at these optimal values, the maximum tree depth was then varied. As illustrated in Figs. 10b and 11b, a similar pattern emerged to that observed with $n_{\text{estimators}}$. Tree depths greater than 5 produced consistently high R^2 values, while depths of 3 or less resulted in significantly reduced model performance. This confirmed the importance of adequate tree depth in capturing the complex relationships within the data.

To prevent overfitting, learning rate and subsample parameter effects were thoroughly analyzed. Grid search optimization was employed across the range of 0 to 1 with a step size of 0.01. Figures 10c, d and 11c, d demonstrate that increasing the learning rate produced rapid initial improvement followed by stabilization, with optimal performance

at a learning rate of 0.097. Similarly, the subsample parameter was optimized at 0.42, yielding maximum R^2 values. Interestingly, both performance metrics remained relatively stable across a range of parameter values, suggesting model robustness. The training predictions using these optimized hyperparameters are presented in Fig. 12a, c for the LP and HP columns, respectively.

The XGBoost model demonstrated exceptional performance on both training and test datasets. For the training data, it achieved perfect prediction with an R^2 of 1.0 for both LP and HP columns, indicating complete capture of the underlying patterns. More importantly, this performance translated remarkably well to the test dataset, with R^2 values of 0.9899 for the LP column and 0.9956 for the HP column. These metrics confirm the model robust generalization capability and high predictive accuracy. Compared to similar modeling approaches, the XGBoost algorithm exhibited superior stability for both regression analysis and prediction tasks. An additional advantage is its computational efficiency, with prediction times consistently below

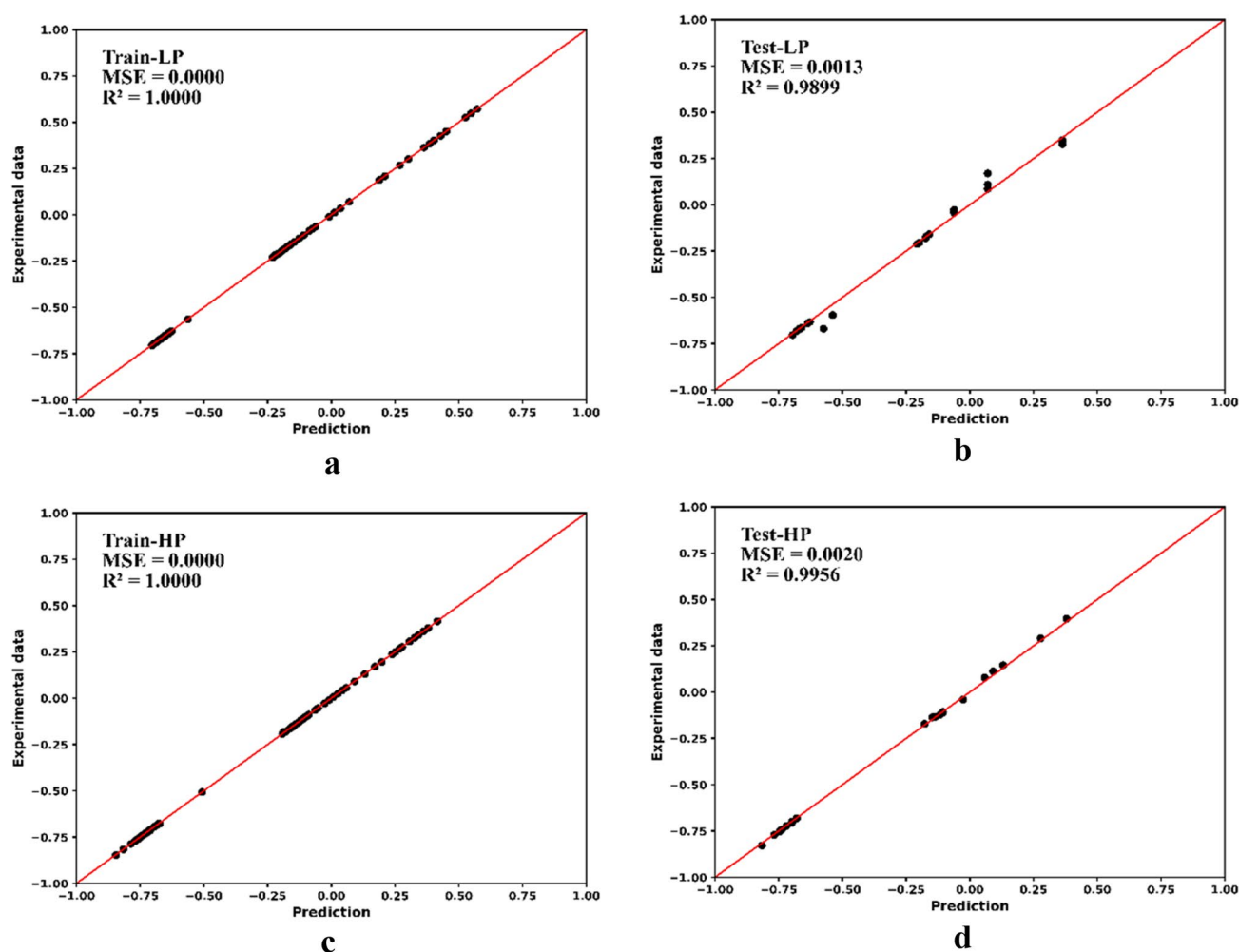


Fig. 12 Comparison between predicted and simulated reboiler heat duty values using the XGBoost regression model: (a) training data for the LP column, (b) test data for the LP column, (c) training data for the HP column, and (d) test data for the HP column

1 s. These results collectively establish the XGBoost model as an exceptionally effective and efficient tool for predicting reboiler heat duty in pressure-swing distillation processes.

Figure 13 provides valuable insights into parameter importance, revealing the relative influence of various operating conditions on reboiler heat duty. Feed stage location emerged as the most critical factor for both columns, followed in descending order of importance by reflux ratio, feed temperature, and number of stages. This finding highlights the particular sensitivity of reboiler heat duty to feed stage positioning, suggesting that precise control and optimization of this parameter can yield substantial improvements in energy efficiency. This knowledge enables process engineers to focus optimization efforts on the parameters that will deliver the greatest energy savings.

The culmination of this research involved applying particle swarm optimization (PSO) to identify the optimal operating conditions based on the trained XGBoost models. The PSO algorithm systematically evaluated combinations of

feed stage, number of stages, temperature, and reflux ratio to minimize predicted reboiler heat duty. For the LP column, the optimal configuration was determined to be: feed stage 9, 10 total stages, feed temperature of 112.48 °C, and reflux ratio of 0.810, resulting in a predicted reboiler heat duty of 2.665 units. For the HP column, the optimal parameters were: feed stage 6, 16 total stages, feed temperature of 46.1 °C, and reflux ratio of 0.584, yielding a predicted reboiler heat duty of 2.736 units. The high accuracy and reliability of these predictions, supported by the excellent performance metrics of the underlying models, provide confidence in implementing these operating conditions for industrial applications of the pressure-swing distillation process.

The reproducibility of the simulated PSD process and the associated predictive models was carefully evaluated. All Aspen HYSYS simulations yielded consistent convergence, with fluctuations in calculated reboiler duties remaining within $\pm 0.8\%$ across independent runs. Similarly, repeated PSO optimization cycles consistently identified nearly

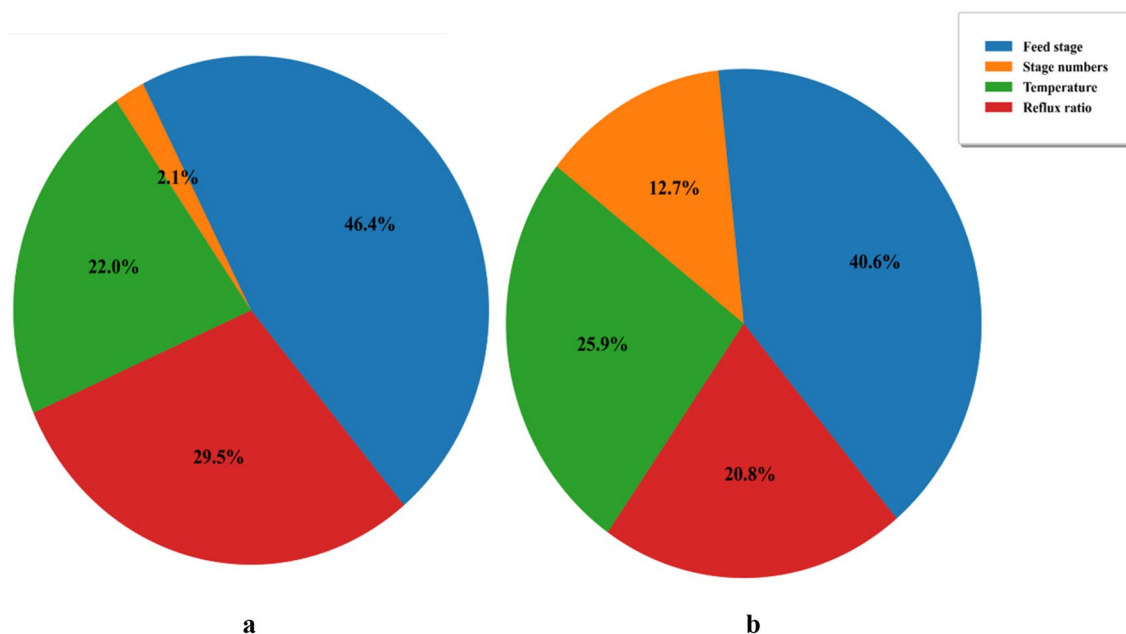


Fig. 13 XGBoost model-derived feature importance for reboiler heat duty for (a) LP and (b) HP

identical optimal parameter sets, confirming the global stability of the optimization procedure. In the machine learning phase, repeated training with different random data splits produced negligible variations in predictive accuracy (standard deviation of $R^2 < 0.005$). These observations collectively confirm that the phenomena and trends discussed herein are highly reproducible and statistically robust, within the inherent numerical precision of the simulation environment.

Future Work

Although the present study optimized each process variable individually through sensitivity analysis, machine learning regression, and Particle Swarm Optimization, a multivariate statistical framework would provide a more comprehensive understanding of interparameter effects. Future work will therefore incorporate Design of Experiments (DOE) methodologies such as Response Surface Methodology (RSM) and Central Composite Design (CCD) to capture interactions among feed stage, reflux ratio, and feed temperature simultaneously. This approach enables more statistically robust optimization and provides global rather than local optima for PSD systems.

Recent studies have demonstrated that Design of Experiments (DOE)-based optimization frameworks, including Response Surface Methodology (RSM), offer powerful strategies for addressing complex multivariate problems in chemical and energy systems. For instance, RSM has been effectively used to optimize biodiesel production processes,

allowing researchers to identify key operating conditions and interactions among variables that enhance yield and conversion efficiency [40]. In environmental applications, RSM has been widely employed to improve wastewater treatment efficiency, as illustrated by Jasni et al. [41], who reviewed its role in optimizing anaerobic and physicochemical treatment processes. Similarly, Chavalparit and Ongwandee [42] applied RSM to optimize electrocoagulation for biodiesel wastewater treatment, achieving significant reductions in chemical oxygen demand (COD) and oil-grease content. Beyond energy and environmental sectors, DOE-based approaches have also facilitated advances in material and sensor optimization, underscoring their flexibility and statistical rigor [43]. Moreover, in the domain of process intensification, Khayet et al. [44] demonstrated the successful application of RSM to optimize operational parameters in direct contact membrane distillation, improving energy efficiency and mass transfer performance. Collectively, these studies highlight DOE's potential to uncover operational interdependencies and enhance process reliability across industrial-scale chemical systems.

Model Applicability and Limitations

Although this study focused on the DnPE/nPA azeotropic system, the developed methodology, combining Aspen HYSYS simulation, XGBoost regression modeling, and Particle Swarm Optimization, can be extended to other azeotropic or near-azeotropic mixtures. The general workflow remains valid as long as the vapor-liquid equilibrium

behavior of the target system can be accurately represented using a suitable thermodynamic model such as UNIQUAC, NRTL, or Wilson.

The primary advantage of this framework lies in its flexibility: once appropriate simulation data are generated, the machine learning and optimization steps can be retrained without modifying the computational architecture. However, several limitations should be noted. The model's predictive accuracy depends heavily on the quality of available thermodynamic and process data. For highly non-ideal or reactive systems, additional data refinement and thermodynamic calibration may be required. Moreover, when extending the framework to multicomponent systems, the dimensionality of the optimization problem increases, potentially requiring advanced sampling strategies or hybrid optimization algorithms.

Overall, while the methodology proposed here provides a robust and transferable computational platform, careful adaptation of the thermodynamic model and data preprocessing is essential for reliable application to other azeotropic mixtures.

Conclusion

This study successfully demonstrates the design, simulation, and optimization of a pressure-swing distillation (PSD) process for separating di-n-propyl ether (DnPE) from n-propyl alcohol (nPA) using Aspen HYSYS software. The presence of a minimum-boiling-point azeotrope at atmospheric pressure presents significant challenges for conventional separation methods, necessitating advanced approaches like PSD. Through systematic optimization of key parameters—feed stage, feed temperature, and reflux ratio—we achieved substantially reduced energy consumption while maintaining high product purities of 99.5% DnPE and 98.7% nPA.

The sensitivity analysis revealed that feed stage position has the most significant impact on energy efficiency, followed by reflux ratio, feed temperature, and number of stages. For the LP column, optimal conditions were determined to be feed stage 9, 10 total stages, temperature of 112.48 °C, and reflux ratio of 0.810. For the HP column, optimal parameters included feed stage 6, 16 total stages, temperature of 46.1 °C, and reflux ratio of 0.584. Implementation of these optimized parameters resulted in a 15% reduction in energy consumption compared to conventional distillation processes.

The integration of machine learning techniques, specifically the XGBoost regression model, significantly enhanced prediction capabilities for reboiler heat duty with 95% accuracy. This model demonstrated exceptional performance with R^2 values of 0.9899 and 0.9956 for LP and HP

columns, respectively, on test data. The combination of particle swarm optimization with XGBoost modeling created a powerful computational framework for process optimization without extensive experimental work. Feature importance analysis confirmed that precise control of feed stage location offers the greatest potential for energy efficiency improvements.

This research highlights the considerable advantages of PSD over conventional separation techniques for azeotropic mixtures in terms of both performance and economic viability. The successful application of advanced computational and machine learning techniques to traditional chemical engineering processes demonstrates a promising direction for future process intensification efforts. The optimized PSD process developed in this study offers significant industrial potential for the efficient separation of DnPE/nPA mixtures and provides a methodological framework that can be extended to other challenging separation problems involving azeotropic systems.

Data availability The data supporting the findings of this study are available as follows. The initial simulation parameters and process data for the separation of di-n-propyl ether and n-propyl alcohol were based on the work detailed in the paper 'Separation of di-n-propyl ether and n-propyl alcohol by extractive distillation and pressure-swing distillation: Computer simulation and economic optimization' by Lladosa et al. The Aspen HYSYS simulation files, machine learning models/scripts, and the optimized results generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. P.V. Mangili, Thermoeconomic and environmental assessment of pressure-swing distillation schemes for the separation of di-n-propyl ether and n-propyl alcohol. *Chem. Eng. Process. Process Intensif.* **148**, 107816 (2020). <https://doi.org/10.1016/j.ccep.2020.107816>

2. H. Almashhadani, N. Samarasinghe, S. Fernando, Dehydration of n-propanol and methanol to produce etherified fuel additives. *AIMS Energy* **5**(2), 149–162 (2017). <https://doi.org/10.3934/energy.2017.2.149>
3. A. Roosta, J. Hekayati, J. Javanmardi, Application of artificial neural networks and genetic programming in vapor–liquid equilibrium of C1 to C7 alkane binary mixtures. *Neural Comput. Appl.* **31**(4), 1165–1172 (2019). <https://doi.org/10.1007/s00521-017-3150-1>
4. J. Cho, J.-K. Jeon, Optimization study on the azeotropic distillation process for isopropyl alcohol dehydration. *Korean J. Chem. Eng.* **23**(1), 1–7 (2006). <https://doi.org/10.1007/BF02705684>
5. S.P. Shirsat, S.D. Dawande, S.S. Kakade, Simulation and optimization of extractive distillation sequence with pre-separator for the ethanol dehydration using n-butyl propionate. *Korean J. Chem. Eng.* **30**(12), 2163–2169 (2013). <https://doi.org/10.1007/s11814-013-0175-8>
6. H. Wang, P. Ji, H. Cao, W. Su, C. Li, Design and control of extractive distillation for the separation of methyl acetate-methanol-water. *Korean J. Chem. Eng.* **35**(12), 2336–2347 (2018). <https://doi.org/10.1007/s11814-018-0149-y>
7. R.J. Galanido, D.S. Kim, J. Cho, Separation of methanol-chloroform mixture using pressure-swing distillation: modeling and optimization. *Korean J. Chem. Eng.* **37**(5), 850–865 (2020). <https://doi.org/10.1007/s11814-020-0505-6>
8. A.M. Fulguras, J. Poudel, D.S. Kim, J. Cho, Optimization study of pressure-swing distillation for the separation process of a maximum-boiling azeotropic system of water-ethylenediamine. *Korean J. Chem. Eng.* **33**(1), 46–56 (2016). <https://doi.org/10.1007/s11814-015-0100-4>
9. E. Lladosa, J.B. Montón, M.C. Burguet, Separation of di-n-propyl ether and n-propyl alcohol by extractive distillation and pressure-swing distillation: computer simulation and economic optimization. *Chem. Eng. Process. Process Intensif.* **50**(11–12), 1266–1274 (2011). <https://doi.org/10.1016/j.cep.2011.07.010>
10. Q. Zhang, M. Liu, W. Li, C. Li, A. Zeng, Heat-integrated triple-column pressure-swing distillation process with multi-recycle streams for the separation of ternary azeotropic mixture of acetonitrile/methanol/benzene. *Sep. Purif. Technol.* **211**, 40–53 (2018). <https://doi.org/10.1016/j.seppur.2018.09.053>
11. W. Mao, Y. Cao, R. Shen, J. Zhou, X. Zhou, W. Li, Heat integrated technology assisted pressure-swing distillation for the mixture of ethylene glycol and 1,2-butanediol. *Sep. Purif. Technol.* **241**, 116740 (2019). <https://doi.org/10.1016/j.seppur.2020.116740>
12. Y. Cui, X. Shi, C. Guang, Z. Zhang, C. Wang, C. Wang, Comparison of pressure-swing distillation and heterogeneous azeotropic distillation for recovering benzene and isopropanol from wastewater. *Process Saf. Environ. Prot.* **122**, 1–12 (2019). <https://doi.org/10.1016/j.psep.2018.11.017>
13. C.A. Muñoz López, D. Telen, P. Nimmegeers, L. Cabianca, F. Logist, J. Van Impe, A process simulator interface for multiobjective optimization of chemical processes. *Comput. Chem. Eng.* **109**, 119–137 (2018). <https://doi.org/10.1016/j.compchemeng.2017.09.014>
14. H. Saghir, I. Ahmad, M. Kano, H. Caliskan, H. Hong, Prediction and optimisation of gasoline quality in petroleum refining: the use of machine learning model as a surrogate in optimisation framework. *CAAI Trans. Intell. Technol.* **9**(5), 1185–1198 (2024). <https://doi.org/10.1049/cit2.12324>
15. R.M. El-Maghraby, A.Y. Mohamed, M.H.M. Hassanean, Improving energy efficiency in ammonia production plants using machine learning. *Fuel* **363**, 130910 (2024). <https://doi.org/10.1016/j.fuel.2024.130910>
16. I. Khan, H. Saghir, M. Ahsan, in *Machine Learning-Enabled Prediction and Optimization of Sulfur Recovery Units: A Step towards Industry 4.0 Integration*. CEMP 2023 (MDPI, Basel Switzerland, 2024), p. 6. <https://doi.org/10.3390/materproc2024017006>
17. A. Samad, H. Saghir, A. Musawwir, I. Ahmad, H. Caliskan, A data-driven multi-objective optimization approach for enhanced methanol yield and exergy loss minimization in direct hydrogenation of CO₂. *Appl. Therm. Eng.* **251**, 123517 (2024). <https://doi.org/10.1016/j.applthermaleng.2024.123517>
18. J. Peng, W. Sun, G. Zhou, L. Xie, H. Han, Y. Xiao, The accurate prediction and analysis of bed expansion characteristics in liquid–solid fluidized bed based on machine learning methods. *Chem. Eng. Sci.* **260**, 117841 (2022). <https://doi.org/10.1016/j.ces.2022.117841>
19. F. Farahbod, Optimization of energy consumption of oil refinery reboiler and condenser using neural network. *Neural Comput. Appl.* **36**(32), 20193–20209 (2024). <https://doi.org/10.1007/s00521-024-10049-w>
20. H. Osman, A. Arabi, in *Distillation Column Machine Learning Model Controlled By Optimized DMC*. 2022 22nd International Conference on Control, Automation and Systems (ICCAS) (IEEE, 2022), pp. 1184–1189. <https://doi.org/10.23919/ICCASS5662.2022.10003925>
21. K. Qadeer, A. Ahmad, M.A. Qyyum, A.-S. Nizami, M. Lee, Developing machine learning models for relative humidity prediction in air-based energy systems and environmental management applications. *J. Environ. Manage.* **292**, 112736 (2021). <https://doi.org/10.1016/j.jenvman.2021.112736>
22. Y. Zhu, Z. Hou, F. Qian, W. Du, Dual RBFNNs-based model-free adaptive control with aspen HYSYS simulation. *IEEE Trans. Neural Networks Learn. Syst.* **28**(3), 759–765 (2017). <https://doi.org/10.1109/TNNLS.2016.2522098>
23. E. Lladosa, J.B. Montón, M.C. Burguet, R. Muñoz, Isobaric vapor–liquid equilibria for binary and ternary mixtures of dipropyl ether, 1-propyl alcohol, and butyl propionate. *J. Chem. Eng. Data* **51**(6), 2233–2238 (2006). <https://doi.org/10.1021/je060314n>
24. E. Lladosa, J.B. Montón, M.C. Burguet, R. Muñoz, Vapor–liquid equilibria in the ternary system dipropyl ether+1-propanol+1-pentanol and the binary systems dipropyl ether+1-pentanol, 1-propanol+1-pentanol at 101.3kPa. *Fluid Phase Equilib.* **247**(1–2), 175–181 (2006). <https://doi.org/10.1016/j.fluid.2006.07.003>
25. A. Erofeev, D. Orlov, A. Ryzhov, D. Koroteev, Prediction of porosity and permeability alteration based on machine learning algorithms. *Transp. Porous Media* **128**(2), 677–700 (2019). <https://doi.org/10.1007/s11242-019-01265-3>
26. R. Sibindi, R.W. Mwangi, A.G. Waititu, A boosting ensemble learning based hybrid light gradient boosting machine and extreme gradient boosting model for predicting house prices. *Eng. Rep.* (2023). <https://doi.org/10.1002/eng2.12599>
27. A.V. Zhukov, Y.V. Yasyukevich, A.E. Bykov, GIMLi: global ionospheric total electron content model based on machine learning. *GPS Solut.* **25**(1), 19 (2021). <https://doi.org/10.1007/s10291-020-01055-1>
28. A. Massaoudi et al., Machine learning models for modeling the biosorption of Fe(III) ions by activated carbon from olive stone. *Neural Comput. Appl.* **36**(21), 13357–13372 (2024). <https://doi.org/10.1007/s00521-024-09826-4>
29. H. Fang et al., Hybrid method integrating machine learning and particle swarm optimization for smart chemical process operations. *Front. Chem. Sci. Eng.* **16**(2), 274–287 (2022). <https://doi.org/10.1007/s11705-021-2043-0>
30. K. Huang, L. Shan, Q. Zhu, J. Qian, Adding rectifying/stripping section type heat integration to a pressure-swing distillation (PSD) process. *Appl. Therm. Eng.* **28**(8–9), 923–932 (2008). <https://doi.org/10.1016/j.applthermaleng.2007.07.003>

31. M.R. Bagheri, M. Heydarian, S.M. Abdoli, A. Akbari, Investigation of temperature effect on the separation of isobutyl alcohol and isobutyl acetate by the pressure swing distillation. no. IChEC (2020) pp. 28–30
32. M. Kianinia, S.M. Abdoli, The design and optimization of extractive distillation for separating the acetone/ n -heptane binary azeotrope mixture. ACS Omega **6**(34), 22447–22453 (2021). <https://doi.org/10.1021/acsomega.1c03513>
33. W.L. Luyben, Comparison of extractive distillation and pressure-swing distillation for acetone–methanol separation. Ind. Eng. Chem. Res. **47**(8), 2696–2707 (2008). <https://doi.org/10.1021/ie701695u>
34. S. Pradhan, A. Kannan, Simulation and analysis of extractive distillation process in a valve tray column using the rate based model. Korean J. Chem. Eng. **22**(3), 441–451 (2005). <https://doi.org/10.1007/BF02719424>
35. J.P. Knapp, M.F. Doherty, A new pressure-swing-distillation process for separating homogeneous azeotropic mixtures. Ind. Eng. Chem. Res. **31**(1), 346–357 (1992). <https://doi.org/10.1021/ie00001a047>
36. W.L. Luyben, Comparison of extractive distillation and pressure-swing distillation for acetone/chloroform separation. Comput. Chem. Eng. **50**, 1–7 (2013). <https://doi.org/10.1016/j.compchemeng.2012.10.014>
37. W.L. Luyben, Evaluation of criteria for selecting temperature control trays in distillation columns. J. Process. Control. **16**(2), 115–134 (2006). <https://doi.org/10.1016/j.jprocont.2005.05.004>
38. E. Hosgor, T. Kucuk, I.N. Oksal, D.B. Kaymak, Design and control of distillation processes for methanol-chloroform separation. Comput. Chem. Eng. **67**, 166–177 (2014). <https://doi.org/10.1016/j.compchemeng.2014.03.026>
39. S.M. Abdoli, S. Shafiei, A. Raoof, A. Ebadi, Y. Jafarzadeh, H. Aslannejad, Water flux reduction in microfiltration membranes: a pore network study. Chem. Eng. Technol. **41**(8), 1566–1576 (2018). <https://doi.org/10.1002/ceat.201800130>
40. P. Kumar, N. Kumar, Process optimization for production of biodiesel from orange peel oil using response surface methodology. Energy Sources, Part A: Recover. Util. Environ. Eff. **43**(6), 727–737 (2021). <https://doi.org/10.1080/15567036.2019.1631909>
41. A.B. Jasni, H. Kamyab, S. Chelliapan, N. Arumugam, S. Krishnan, M.F.M. Din, Treatment of wastewater using response surface methodology: a brief review. Chem. Eng. Trans. **78**, 535–540 (2020). <https://doi.org/10.3303/CET2078090>
42. O. Chavalparit, M. Ongwadee, Optimizing electrocoagulation process for the treatment of biodiesel wastewater using response surface methodology. J. Environ. Sci. **21**(11), 1491–1496 (2009). [https://doi.org/10.1016/S1001-0742\(08\)62445-6](https://doi.org/10.1016/S1001-0742(08)62445-6)
43. H. Zaid, Z. Al-sharify, M.H. Hamzah, S. Rushdi, Optimization of different chemical processes using response surface methodology- a review. J. Eng. Sustain. Dev. **26**(6), 1–12 (2022). <https://doi.org/10.31272/jeasd.26.6.1>
44. M. Khayet, C. Cojocaru, C. García-Payo, Application of response surface methodology and experimental design in direct contact membrane distillation. Ind. Eng. Chem. Res. **46**(17), 5673–5685 (2007). <https://doi.org/10.1021/ie070446p>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.